Steps: One Website on Multiple Ports (Advanced)

Condensed, ordered command list. Full explanations are in README.md; fixes for problems are in TROUBLESHOOTING.md.

Step 1 — Install Nginx

```
        sudo apt update && sudo apt install nginx -y && sudo systemctl
enable --now nginx   # Ubuntu/Debian
        sudo dnf install nginx -y && sudo systemctl enable --now nginx
# CentOS/RHEL
```

Step 2 — Create Shared Site Content + Health Endpoint

```
        sudo mkdir -p /var/www/mysite/health
        sudo tee /var/www/mysite/index.html > /dev/null << 'EOF'
        <h1>My Website</h1><p>Port: <span id="port"></span></p>
        <script>document.getElementById('port').textContent =
window.location.port || '80';</script>
        EOF
        echo "OK" | sudo tee /var/www/mysite/health/index.html
        sudo chown -R www-data:www-data /var/www/mysite     # Ubuntu/Debian
        # sudo chown -R nginx:nginx /var/www/mysite         # CentOS/RHEL
```

Step 3 — Write the Multi-Port Config

Create `/etc/nginx/sites-available/mysite` (Ubuntu/Debian) or `/etc/nginx/conf.d/mysite.conf` (CentOS/RHEL):

```
log_format multiport '$remote_addr - $server_port [$time_local]
"$request" $status $body_bytes_sent rt=$request_time';

server {
    listen 8081;
    listen 8082;
    server_name localhost;
    root /var/www/mysite;
    index index.html;

    access_log /var/log/nginx/mysite_access.log multiport;
    error_log  /var/log/nginx/mysite_error.log warn;

    add_header X-Frame-Options "SAMEORIGIN" always;
    add_header X-Content-Type-Options "nosniff" always;
    add_header X-Served-Port $server_port always;

    gzip on;
    gzip_types text/plain text/css application/javascript
application/json text/xml;
```

```
    location / { try_files $uri $uri/ =404; }
    location ~ /\. { deny all; }
}

server {
    listen 8083;
    server_name localhost;
    root /var/www/mysite;
    access_log off;

    location = /health {
        default_type text/plain;
        try_files /health/index.html =503;
    }
    location / { return 404; }
}
```

Step 4 — Enable the Site (Ubuntu/Debian only)

```
sudo ln -s /etc/nginx/sites-available/mysite /etc/nginx/sites-enabled/
```

CentOS/RHEL: skip — `conf.d/` loads automatically.

Step 5 — Test and Reload

```
sudo nginx -t
sudo systemctl reload nginx
```

Step 6 — Open Firewall Ports

```
sudo ufw allow 8081/tcp
sudo ufw allow 8082/tcp
sudo ufw allow from 10.0.0.0/8 to any port 8083 proto tcp

sudo firewall-cmd --permanent --add-port=8081/tcp
sudo firewall-cmd --permanent --add-port=8082/tcp
sudo firewall-cmd --permanent --zone=internal --add-port=8083/tcp
sudo firewall-cmd --reload
```

SELinux (CentOS/RHEL, enforcing):

```
sudo semanage port -a -t http_port_t -p tcp 8081
sudo semanage port -a -t http_port_t -p tcp 8082
sudo semanage port -a -t http_port_t -p tcp 8083
```

Step 7 — Verify

```
curl -s http://localhost:8081 | md5sum
curl -s http://localhost:8082 | md5sum
curl -I http://localhost:8081 | grep X-Served-Port
curl http://localhost:8083/health
curl -i http://localhost:8083/
```

Step 8 — (Optional) Watch Logs Live

```
sudo tail -f /var/log/nginx/mysite_access.log
```

Step 9 — (Optional) Turn Port 8082 into a Redirect Instead of a Mirror

```
server {
    listen 8082;
    server_name localhost;
    return 301 http://$host:8081$request_uri;
}
```

Then re-run Step 5.